

Exploring Auxiliary Objectives through Multi-Task Learning for Small-Data Language Model Pretraining: A BabyLM 2026 Submission

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Abstract

Pretraining optimizes a language model with self-supervised objectives over raw text, shaping the representations later measured by diagnostic evaluations. Masked language modeling (MLM) predicts hidden tokens from context, combining the distributional hypothesis with the cloze task. Less is known about alternative objectives under small data budgets. Using BabyLM 2026’s 10M-token Strict-Small setting, this paper asks whether auxiliary objectives jointly improve grammar, world-knowledge plausibility, entity tracking, and reading prediction, or instead benefit some areas at the expense of others. The experiments compare an MLM-only baseline with several multi-task objective mixtures while holding the corpus, architecture, tokenizer, training budget, and evaluation pipeline fixed. Results are selective: MLM-only is strongest on grammatical minimal-pair judgments, while auxiliary objectives improve entity-state tracking and reading prediction. Within the controlled fast evaluation, single-mask semantic cloze gives the clearest directional world-knowledge plausibility gain; grammar repair does not recover grammatical performance, and broad plausibility classification produces no positive EWoK delta. Auxiliary objectives therefore act as targeted steering signals rather than uniform improvements over MLM.

1 Introduction

The distributional hypothesis motivates pretraining by characterizing linguistic units through context (Harris, 1954). Masked language modeling (MLM) operationalizes this idea by recovering hidden tokens (Devlin et al., 2019), a modern form of the cloze task (Taylor, 1953). Alternative cloze objectives can target function words, discourse connectives, agreement, semantic affordances, and plausibility. Under small data budgets, objective choice determines which contrasts a compact model sees

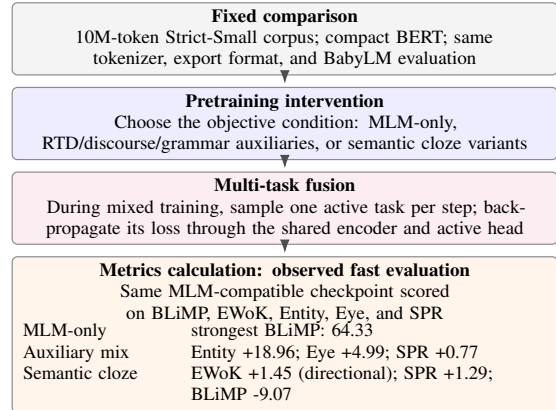


Figure 1: Thesis diagram. The experiment keeps the corpus, architecture, tokenizer, and evaluation interface fixed, varies the pretraining objective mixture, fuses tasks by sampling one active loss per update through the shared encoder, and computes the same BabyLM fast diagnostics. Deltas are representative fast-evaluation changes from the MLM-only baseline; the EWoK change is interpreted as directional.

often enough to learn, yet these effects remain underexplored.

Small-data pretraining matters when web-scale corpora or compute are unavailable, including for low-resource languages (Joshi et al., 2020), specialized domains (Gururangan et al., 2020), and on-device deployment (Lu et al., 2025). BabyLM formalizes this concern by evaluating language models trained under restricted, human-scale data budgets (Warstadt et al., 2023). Its premise is that progress should include closing the data-efficiency gap between human and computational learners, not only scaling data. The 2026 English release provides a 100M-token *Strict* corpus and a 10M-token *Strict-Small* corpus; this paper uses *Strict-Small* throughout.¹ The fixed 10M-token budget makes sample efficiency the experimental target and prevents improvements from being attributed to additional text.

¹<https://babylm.github.io/>

Prior BabyLM submissions explore architectures, data filtering, curricula, and optimization for sample-efficient pretraining (Warstadt et al., 2023; Haller et al., 2024). This study instead holds the corpus, architecture, tokenizer, training budget, and evaluation interface fixed while varying the objective mixture. A compact 11.2M-parameter BERT-style model is trained without downstream fine-tuning under MLM-only or multi-task mixtures containing replacement detection, discourse and function-word prediction, grammatical contrasts, semantic plausibility, and cloze-style semantic ranking. The objective mixture is therefore the main intervention. The design asks: *Under the BabyLM Strict-Small data budget, how do auxiliary pretraining objectives change diagnostic performance relative to MLM-only pretraining?* (RQ)

The answer is selective improvement. Auxiliary objectives improve entity tracking and reading prediction, and semantic cloze gives the clearest directional world-knowledge plausibility gain in the controlled fast evaluation, but MLM-only remains strongest on grammatical minimal pairs. Targeted tasks also transfer unevenly: grammar repair does not recover grammatical performance, while broad plausibility classification produces no positive fast-evaluation EWoK delta. The contribution is therefore a controlled account of how auxiliary objectives steer a compact model toward behaviors aligned with their loss and prediction interface rather than uniformly improving MLM. Figure 1 summarizes this thesis.

2 Background and Objective Families

Within BabyLM Strict-Small, the pretraining corpus is limited to 10M tokens. This creates pressure to choose not only model architecture and data processing, but also the pretraining signal. The baseline objective in this study is MLM, as used by BERT-style encoders (Devlin et al., 2019). MLM trains a model to reconstruct masked tokens from context and remains a strong default for compact encoder-only models.

Auxiliary objectives are attractive because they can make specific contrasts more frequent than they would be under random token masking. Replaced-token detection, for example, gives dense corruption-discrimination supervision related to ELECTRA-style pretraining (Clark et al., 2020). Discourse and function-word tasks can highlight connectives and determiners. Pairwise objectives

can ask the model to prefer natural collocations, grammatical sentences, or semantically plausible contexts. These signals may help diagnostics that depend on discourse tracking, reading alignment, or conceptual plausibility.

The risk is that auxiliary supervision may interfere with MLM. Some tasks use classifier heads rather than the MLM head used by evaluation, and even MLM-compatible tasks can change which behaviors are emphasized during training. For this reason, aggregate scores are less informative than the pattern of gains and losses across benchmarks. The analysis therefore treats each objective family as a possible steering signal for different diagnostic behaviors.

3 Experimental Setup

Pretraining corpus. All experiments use the official BabyLM 2026 Strict-Small training set, `BabyLM-community/BabyLM-2026-Strict-Small`.² The dataset card describes it as a detoxified 10M-token English training set and reports six source files: BNC spoken, CHILDES, Gutenberg, OpenSubtitles, Simple Wikipedia, and Switchboard. In the training code, examples are loaded from the `text` column, empty rows are filtered, and the remaining rows are split with a fixed seed into a training portion and 2,000 held-out validation examples.

Tokenization and validation. All systems use the same `bert-base-uncased` WordPiece tokenizer. After tokenization, the text is concatenated and grouped into fixed-length sequences of 128 tokens. The held-out corpus split is used for validation loss, checkpoint selection, early stopping, and MLM calibration diagnostics; it is not used as a reported BabyLM benchmark score.

Auxiliary-data setup. The auxiliary datasets are built only after fixing this corpus split and model interface. Corpus-derived objectives reuse the Strict-Small text and statistics from it, while template-derived objectives instantiate controlled grammar, plausibility, semantic-cloze, and relational contrasts. Official BabyLM diagnostic items remain evaluation-only: the training procedure does not use BLiMP, BLiMP Supplement, EWoK, Entity Tracking, or reading-prediction benchmark examples. This setup keeps the underlying data source

²<https://huggingface.co/datasets/BabyLM-community/BabyLM-2026-Strict-Small>

fixed while varying how the same small-data setting is converted into pretraining supervision.

3.1 Auxiliary objective generation

Table 1 summarizes the objective families tested. All systems keep the same compact BERT-style architecture and tokenizer. Most auxiliary data are generated from the BabyLM training corpus or from small synthetic templates. No official BLiMP, EWoK, Entity Tracking, or reading-prediction items are used to construct the auxiliary tasks.

Generation. Corpus-derived objectives are generated offline from the BabyLM Strict-Small training split. The generator builds token and bi-gram frequency statistics, masks eligible tokens for MLM, creates RTD and collocation negatives by replacing words with alternatives from the same coarse word class and frequency bucket, and creates connective and definiteness examples by replacing a fixed connective inventory or articles with special markers. The syntax-repair tasks use conservative local filters to mask function words and agreement-bearing tokens. The grammar, conceptual-plausibility, semantic-cloze, and relational-cloze objectives are template-based: templates instantiate grammatical contrasts, plausible versus implausible event roles, affordances, part-whole relations, physical properties, typical locations, and spatial or relational contrasts. For pairwise tasks, the plausible or grammatical option is randomly placed first or second, and the label records which side should be preferred.

Objective families. The used tasks are grouped into five families, with detailed per-task examples in Appendix A.

1. **Corruption and discrimination objectives** turn contextual fit into a reconstruction or anomaly-detection problem, following the distributional and cloze motivation behind MLM (Harris, 1954; Taylor, 1953; Devlin et al., 2019) and RTD-style pretraining (Clark et al., 2020). For example, MLM predicts *apple* in “The child ate the [MASK],” while RTD marks *sky* as corrupted in “The cat sat on the sky.”
2. **Discourse and function objectives** target closed-class words whose form is tied to discourse relations, reference, or syntactic structure (Halliday and Hasan, 1976; Prasad et al.,

2008; Gundel et al., 1993; Fries, 1952). For example, connective prediction recovers *so* in “It was raining, [unused1] the family stayed inside.”

3. **Grammatical repair objectives** emphasize agreement and controlled acceptability contrasts (Corbett, 2006; Chomsky, 1957; Warstadt et al., 2020). For example, an agreement item predicts *are* in “The dogs [MASK] running,” and a minimal-pair item prefers “The dogs run” over “The dogs runs.”
4. **Semantic plausibility objectives** train pairwise choices over plausible and implausible contexts, motivated by selectional preferences and world-knowledge plausibility (Resnik, 1996; Ivanova et al., 2025). For example, a classifier prefers the context “The apple was a fruit” over “The apple was a stone” for “The child ate the apple.”
5. **MLM-head cloze and ranking objectives** keep semantic supervision close to the masked-LM scoring interface by ranking plausible completions at masked positions (Salazar et al., 2020). For example, semantic cloze ranks *fly* over *read* in “A bird can [MASK],” while relational cloze ranks *on* over *under* in “The picture is [MASK] the wall.”

The semantic and relational cloze objectives use a masked-token ranking loss. For these examples, let s_g and s_b be the MLM log-probabilities of the plausible and implausible tokens at the masked position. The ranking loss is

$$\mathcal{L}_{cloze} = \log(1 + \exp(-(s_g - s_b))). \quad (1)$$

This objective changes pretraining while preserving a normal `AutoModelForMaskedLM` evaluation interface.

3.2 Training and evaluation protocol

Figure 2 summarizes the experimental control structure. The fixed layers define the comparison space: the corpus, model family, tokenizer, training budget, checkpoint interface, and evaluation suite remain constant. The objective-condition layer is the intervention, so score differences are interpreted as effects of pretraining objective mixture rather than changes in architecture or evaluation protocol.

Family	Task examples	Training signal	Expected target	Observed effect
MLM-only	Random masked tokens	MLM reconstruction	General baseline; BLiMP	Strongest BLiMP; weakest Entity and reading scores.
Core multi-task	RTD, connective, definiteness, collocation	Token and sequence classification plus MLM	Entity, reading, discourse/function words	Large Entity and reading gains, large BLiMP loss.
Syntax repair	Function-word recovery, agreement prediction	MLM-head recovery of selected syntax tokens	BLiMP recovery	Only small BLiMP recovery; Entity and reading remain strong.
Grammar pair repair	BLiMP-style generated minimal pairs	Pairwise classifier over sentence pairs	BLiMP recovery	Does not recover BLiMP; weak positive fast-evaluation EWoK delta.
Conceptual plausibility	Selectional, affordance, containment, part-whole, agency templates	Pairwise plausibility classifier	EWoK and Entity	Improves Entity, but has a negative fast-evaluation EWoK delta.
MLM-head pair ranking	Plausible vs. implausible contexts from the same templates	MLM-head ranking over masked context tokens	EWoK and reading	Improves reading most strongly, but has a negative fast-evaluation EWoK delta.
Semantic cloze	Single-mask plausible vs. implausible completion	MLM-head ranking at the masked position	EWoK	Clearest directional fast-evaluation EWoK gain; still trades off against BLiMP.
Relational cloze	Spatial and relational masked-token contrasts	MLM-head ranking at the masked position	EWoK and Entity	Best Entity score, but lower fast-evaluation EWoK than semantic cloze 5% final.

Table 1: Auxiliary objective families tested in the BabyLM Strict-Small experiments.

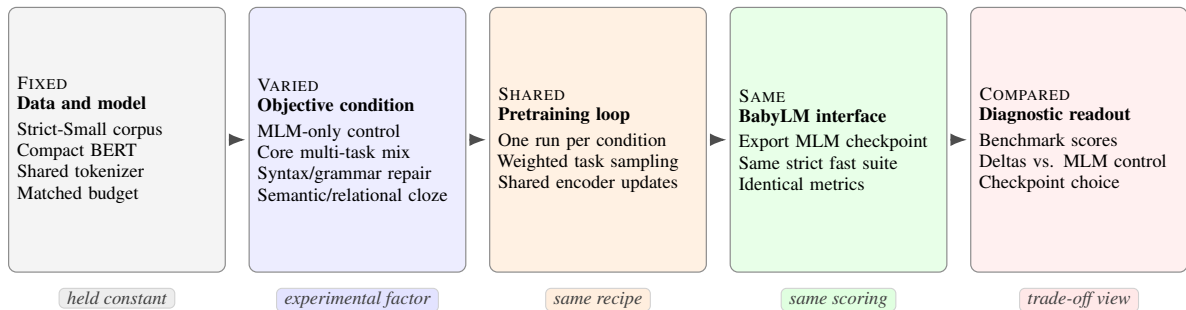


Figure 2: Experimental setup. All conditions share the Strict-Small corpus, compact BERT architecture, tokenizer, training budget, checkpoint interface, and BabyLM strict fast-evaluation suite. The only planned intervention is the objective condition: the MLM-only control is compared with auxiliary multi-task, syntax/grammar repair, semantic cloze, and relational cloze variants. Results are interpreted as benchmark-level deltas from the MLM-only control.

Model and data control. All comparisons use a compact BERT-style masked language model with hidden size 256, 4 transformer layers, 4 attention heads, intermediate size 1024, maximum position embeddings 512, and the `bert-base-uncased` WordPiece tokenizer. The exported semantic-cloze checkpoint has 11,201,338 parameters. All runs use the shared corpus and preprocessing described in Section 3; no condition changes the underlying BabyLM Strict-Small data source.

How separate tasks form one model. The tasks are not collapsed into one label space, and separately trained models are not ensembled. Instead, they are integrated by alternating updates to one shared encoder:

1. **Keep task data separate.** Each objective has its own JSONL file, data loader, sampling probability, target format, and loss.
2. **Select one task per step.** The trainer samples

a task from the configured mixture and draws one batch only from that task. Losses from different tasks are therefore not averaged on the same batch.

3. **Update shared parameters.** The batch passes through the same token embeddings and BERT encoder, then through the head required by the selected task. Backpropagation updates the shared encoder and active head; inactive heads receive no update on that step.
4. **Export one model.** After many alternating steps, the shared encoder has accumulated updates from every sampled objective. Auxiliary heads are removed, leaving that encoder and the MLM head as one standard `AutoModelForMaskedLM` checkpoint.

The shared encoder is therefore the point of multi-task fusion: task knowledge is combined in one parameter set across training steps, not by combining predictions at evaluation time.

Task-to-head routing. MLM, function-word recovery, agreement prediction, MLM-head pair ranking, semantic cloze, and relational cloze train through the MLM head. RTD uses a token classifier; connective and definiteness prediction use marker-position classifiers; and collocation, grammar minimal-pair, and conceptual plausibility tasks use a two-way classifier over $[\text{CLS}]$. These auxiliary classifiers are training scaffolds and are not included in the submitted checkpoint. The selected semantic-cloze run alternates MLM, RTD, connective, definiteness, collocation, grammar minimal-pair, and semantic-cloze batches; semantic cloze has a configured 5% probability during the mixed-task phase, followed by an MLM-only calibration phase. The semantic cloze 5% final checkpoint is the single official submission artifact; other checkpoints are diagnostic comparisons rather than separate submitted systems. Appendix B and Figure 3 visualize this update-and-export process.

Training. Unless otherwise noted, runs use batch size 96, maximum sequence length 128, AdamW with learning rate 5×10^{-4} , weight decay 0.01, linear warmup for 6% of steps, gradient clipping at 1.0, fp32 training, and a 10,000-step budget. Several multi-task runs use a final MLM-only calibration phase or early stopping after step 9000. The semantic cloze 5% run stops at step 9500; its best validation checkpoint occurs at step 5000, while its final checkpoint is the strongest fast-evaluation EWoK checkpoint.

Evaluation benchmarks and metrics. The empirical analysis reports verified local outputs from the official BabyLM strict fast-evaluation suite for each experimental checkpoint.³ The fast suite is a subsampled diagnostic evaluation used for checkpoint comparison (Warstadt et al., 2023). For candidate-ranking benchmarks, let $s(c_{ij})$ be the summed token log-probability assigned by the MLM backend to candidate j for item i , and let y_i be the correct candidate. Accuracy is $100N^{-1} \sum_i \mathbb{I}[\arg \max_j s(c_{ij}) = y_i]$. The reported benchmarks are:

1. **BLiMP: grammatical minimal pairs.** Each item contrasts a grammatical sentence with a minimally different ungrammatical alternative (Warstadt et al., 2020). The metric is the percentage of pairs for which the grammatical sentence receives the higher MLM score.

2. **BLiMP Supplement: additional controlled linguistic contrasts.** The supplement extends minimal-pair evaluation to hypernymy, question-answer congruence, subject-auxiliary inversion, and turn-taking phenomena used in BabyLM (Warstadt et al., 2023). It uses the same candidate-ranking accuracy calculation as BLiMP.
3. **EWoK: conceptual and world-knowledge plausibility.** Each item asks the model to prefer the completion supported by an everyday-world context over a controlled alternative (Ivanova et al., 2025). The metric is candidate-ranking accuracy, aggregated by the official pipeline across EWoK domains.
4. **Entity Tracking: discourse-state consistency.** Items describe sequences of entity-state changes and test whether the model prefers the continuation that correctly identifies the resulting state (Kim and Schuster, 2023). The BabyLM adaptation reports candidate-ranking accuracy rather than free-form generation accuracy.
5. **Eye Tracking: alignment with human gaze durations.** Model surprisal is added to a regression baseline containing lexical and contextual controls for four eye-tracking measures (de Varda et al., 2024). The score is the mean normalized incremental fit, $100(R_{\text{LM}}^2 - R_{\text{base}}^2)/(1 - R_{\text{base}}^2)$; higher values mean that surprisal explains more residual reading-time variance.
6. **Self-Paced Reading (SPR): alignment with human reading times.** The same normalized incremental- R^2 score is computed for self-paced reading time, with the official regression accounting for previous-word spillover (Frank et al., 2013). Higher values indicate greater incremental predictive power from current-word surprisal.

All comparisons keep the model architecture, tokenizer, corpus, and evaluation pipeline fixed. Table 2 summarizes the objective mixtures behind the reported rows. Rows labeled *best* use the held-out MLM-validation checkpoint unless otherwise stated; rows labeled *final* use the terminal or early-stopped checkpoint. The best-by-metric table is a retrospective diagnostic readout, not a checkpoint-selection rule. The controlled comparisons use fast evaluation throughout. Official full evaluation of

³<https://github.com/babylm-org/babylm-eval>

the selected submission yields 50.45 EWoK accuracy, but no matched full-evaluation baseline is used; this value is therefore descriptive and is not treated as evidence of an EWoK improvement.

4 Results

MLM-only is compared against auxiliary multi-task training across all diagnostic families. Two supporting analyses help explain the observed changes: comparisons among classifier-style and MLM-head objectives test whether transfer depends on objective format, while comparisons between best-validation and final checkpoints examine whether targeted gains persist across training. These analyses are diagnostic and are not estimates of robust checkpoint-selection performance.

Table 3 gives the raw fast-evaluation scores for representative checkpoints. The main pattern is selective improvement. The strongest BLiMP result is the MLM-only baseline at 64.33. Every auxiliary system loses at least 7.88 BLiMP points relative to that baseline, including objectives explicitly designed to repair grammatical preference. This makes BLiMP the clearest cost of auxiliary training.

At the same time, the auxiliary systems produce large gains for Entity Tracking and reading. The calibrated multi-task system improves Entity Tracking from 15.56 to 34.52, Eye from 2.47 to 7.46, and SPR from 2.42 to 3.19. Most later auxiliary variants preserve these gains. The best Entity score is achieved by the relational cloze best checkpoint, 40.68, and the best Eye score by the 2.5% semantic cloze final checkpoint, 8.82.

Within the controlled fast evaluation, EWoK is more selective. The semantic results show that task format matters more than broad semantic motivation. The conceptual plausibility classifier and MLM-head pair-ranking objective are both motivated by EWoK-style plausibility, yet neither produces a positive fast-evaluation EWoK delta. In contrast, semantic cloze ranking directly trains the MLM head to prefer plausible single-token completions over implausible alternatives, matching the masked-token evaluation interface more closely. The 5% semantic cloze final checkpoint reaches 52.00 in fast evaluation, 1.45 points above the fast-evaluation MLM baseline and 1.00 above the previous best non-cloze result. Because these are single-seed runs and EWoK scores remain close to chance, this fast-evaluation difference is interpreted

as a directional effect rather than a stable estimate of expected improvement. More generally, checkpoint comparisons are used diagnostically rather than as estimates of expected generalization. The semantic-cloze weight sweep and relational variant do not beat this result, but they clarify the trade-off. Increasing semantic cloze to 10% gives the strongest non-selected sweep EWoK score of 51.64 in fast evaluation. Relational cloze does not beat semantic cloze on fast-evaluation EWoK, but it produces the strongest Entity Tracking score, suggesting that relation-focused masked-token supervision may strengthen entity-state or discourse tracking more than the kind of conceptual plausibility measured by EWoK.

Table 4 shows the same pattern as deltas from the MLM baseline. The consistent sign structure is the main finding: fast-evaluation BLiMP deltas are negative for every auxiliary system, Entity and reading deltas are strongly positive for nearly every auxiliary system, and the EWoK delta becomes positive only for cloze-shaped objectives. Thus, the central result is that auxiliary objectives move gains toward particular diagnostic families.

To address **RQ**, the results show that auxiliary objectives selectively improve diagnostic performance rather than producing a uniform overall gain. They substantially improve Entity Tracking and reading-aligned metrics, while semantic cloze produces a directional positive EWoK delta in fast evaluation; however, every auxiliary system reduces BLiMP accuracy relative to MLM-only. The supporting comparisons suggest that objective format helps explain this pattern: grammar minimal-pair classification does not recover BLiMP, conceptual-plausibility classification does not produce a positive EWoK delta, and the strongest fast-evaluation EWoK result appears when semantic supervision is expressed as masked-token cloze ranking through the MLM head.

5 Analysis

Objective/evaluation alignment. The main conclusion is not that auxiliary objectives simply help or hurt. They reallocate a small pretrained model toward the decisions emphasized during training. Under a fixed small-data budget, auxiliary objectives are therefore useful steering signals, but they are not neutral extra supervision. They consume updates that would otherwise train ordinary MLM, and they shape the representation through the in-

Condition	Active objective mixture	Reported checkpoint(s)	Selection note
MLM-only	Random MLM only	final	Baseline
Calibrated multi-task	MLM .60, RTD .20, connective .10, definiteness .10	validation-best	AutoModelForMaskedLM. Final 2k steps use MLM-only calibration.
Syntax repair	MLM .60, RTD .10, connective .075, definiteness .075, collocation .05, function-word .05, agreement .05	final; best discussed	Final checkpoint has strongest BLiMP/reading for this run.
BLiMP-pair repair	MLM .60, RTD .10, connective .075, definiteness .075, collocation .05, grammar-minpair .10	validation-best	Final 2k steps use MLM-only calibration.
Conceptual plausibility	MLM .55, RTD .10, connective .075, definiteness .075, collocation .05, conceptual-plausibility .15	validation-best	Classifier-style semantic objective.
MLM-head pair ranking	Same base mixture as conceptual plausibility, replacing the classifier task with MLM-head pair ranking	validation-best	MLM-compatible ranking over plausible/implausible contexts.
Semantic cloze 5%	MLM .60, RTD .10, connective .075, definiteness .075, collocation .05, grammar-minpair .05, semantic-cloze .05	validation-best and final	Final checkpoint gives the strongest observed fast-evaluation EWoK and SPR.
Semantic cloze sweep	Same base as semantic cloze, with cloze weight .025, .075, or .10 and MLM weight adjusted inversely	selected final/best rows	Used to test whether the 5% result is weight-specific.
Relational cloze	MLM .60, RTD .10, connective .075, definiteness .075, collocation .05, grammar-minpair .05, semantic-cloze .025, relational-cloze .025	validation-best and final	Tests whether relation-focused cloze transfers to EWoK or Entity.

Table 2: Configuration summary for reported systems. Probabilities are task-sampling weights during mixed training. Rows marked validation-best use the held-out MLM-validation checkpoint, not the BabyLM diagnostic score.

System	BLiMP	Sup.	EWoK	Entity	Eye	SPR	Main takeaway
MLM-only baseline	64.33	54.00	50.55	15.56	2.47	2.42	strongest BLiMP
Calibrated multi-task	54.80	54.80	50.27	34.52	7.46	3.19	Entity and reading gains
Syntax repair final	55.51	52.40	49.55	32.76	8.28	3.63	repair does not recover BLiMP
BLiMP-pair repair best	54.80	51.60	51.00	27.54	8.02	3.05	weak positive fast EWoK delta
Conceptual plausibility best	55.46	56.40	49.64	37.10	7.87	3.22	high Entity, negative EWoK delta
MLM pair ranking best	55.95	47.20	49.55	33.50	8.73	3.67	strong reading, negative EWoK delta
Semantic cloze 5% best	54.10	57.60	50.73	36.61	7.56	3.11	high Supplement/Entity
Semantic cloze 5% final	55.26	54.40	52.00	27.44	8.18	3.71	best fast EWoK and SPR
Semantic cloze 2.5% final	54.34	54.80	50.73	22.79	8.82	3.06	best Eye
Semantic cloze 7.5% best	53.57	52.80	51.00	40.48	7.85	3.13	high Entity
Semantic cloze 10% final	55.09	52.80	51.64	33.59	8.01	3.39	best non-selected sweep EWoK
Relational cloze best	54.78	53.20	51.00	40.68	7.69	3.29	best Entity
Relational cloze final	54.83	52.00	50.82	26.80	8.09	3.26	relation final lowers Entity

Table 3: Official strict fast-evaluation results for representative systems. BLiMP, Supplement, EWoK, and Entity Tracking are accuracy-style percentages. Eye and SPR are official reading-prediction scores.

terface that receives the loss. This makes objective design a question of alignment: an auxiliary task is most useful when its loss, prediction head, and contrast structure resemble the diagnostic behavior one hopes to improve. The strongest gains appear when the auxiliary training signal resembles the diagnostic mechanism used at evaluation time. Semantic cloze is associated with the clearest directional fast-evaluation EWoK gain and uses the MLM head with a masked-token plausibility contrast. MLM-head pair ranking improves reading-oriented scores, plausibly by changing token-level expectations, but does not by itself produce a positive fast-evaluation EWoK delta. Conceptual plausibility improves Entity Tracking but not fast-evaluation EWoK, suggesting that a pairwise classifier over contexts may not transfer to the masked-LM scoring interface.

Targeted objectives do not always transfer. The non-transfer cases are as informative as the

gains. Function-word and agreement repair improve BLiMP only slightly. A direct BLiMP-style grammatical minimal-pair classifier also does not recover BLiMP, suggesting that template pair classification is not enough to shape the sentence pseudo-likelihood preferences used by BLiMP scoring. Similarly, broad conceptual plausibility supervision shows no positive fast-evaluation EWoK delta unless it is recast as a single-mask cloze ranking objective. The format of the objective matters, not just its linguistic label.

Checkpoint dynamics. The best validation checkpoint is not necessarily the best diagnostic checkpoint. In semantic cloze, the best-validation checkpoint has higher BLiMP Supplement and Entity Tracking than the final checkpoint, but the final checkpoint has higher fast-evaluation EWoK and SPR. In the relational cloze run, the best checkpoint reaches Entity 40.68 while the final check-

System	Δ BLiMP	Δ Sup.	Δ EWoK	Δ Entity	Δ Eye	Δ SPR
Calibrated multi-task	-9.53	+0.80	-0.28	+18.96	+4.99	+0.77
Syntax repair final	-8.82	-1.60	-1.00	+17.20	+5.81	+1.21
BLiMP-pair repair best	-9.53	-2.40	+0.45	+11.98	+5.55	+0.63
Conceptual plausibility best	-8.87	+2.40	-0.91	+21.54	+5.40	+0.80
MLM pair ranking best	-8.38	-6.80	-1.00	+17.94	+6.26	+1.25
Semantic cloze 5% final	-9.07	+0.40	+1.45	+11.88	+5.71	+1.29
Semantic cloze 10% final	-9.24	-1.20	+1.09	+18.03	+5.54	+0.97
Relational cloze best	-9.55	-0.80	+0.45	+25.12	+5.22	+0.87

Table 4: Representative fast-evaluation deltas from the MLM-only baseline. Positive values indicate higher fast-evaluation scores than MLM-only; EWoK deltas are interpreted directionally.

Metric	Best system	Score
BLiMP	MLM-only baseline	64.33
Supplement	Semantic cloze 5% best	57.60
EWoK	Semantic cloze 5% final	52.00
Entity	Relational cloze best	40.68
Eye	Semantic cloze 2.5% final	8.82
SPR	Semantic cloze 5% final	3.71

Table 5: Best observed fast-evaluation system by metric. No single checkpoint dominates across the diagnostic suite.

point drops to 26.80. These shifts suggest that BabyLM-scale models may pass through different diagnostic regimes during training, even under the same objective mixture.

6 Conclusion

Auxiliary objectives in BabyLM Strict-Small are useful, but their benefit is selective. Under a fixed small corpus and compact BERT-style architecture, they improve several non-BLiMP diagnostics: multi-task objectives substantially improve Entity Tracking and reading-aligned metrics, and direct semantic cloze ranking produces the clearest directional EWoK gain in the controlled fast evaluation. The main cost is grammatical minimal-pair performance, where MLM-only remains strongest on BLiMP. Thus, the strongest conclusion is a trade-off: auxiliary objectives do not uniformly improve the model, but they can steer small-data pretraining toward the diagnostic behaviors their losses emphasize. The broader lesson is that auxiliary objectives should be designed and evaluated with their target diagnostic interface in mind, because pairwise classifiers, MLM-head ranking, and single-mask cloze losses do not transfer in the same way. Future work should test whether these selective gains and BLiMP trade-offs persist across seeds, larger models, and more systematic objective-weight schedules.

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A Detailed Task Examples

This appendix expands the task inventory summarized in the main text. The citations identify the linguistic or psycholinguistic constructs that inspired each generated objective; they are not sources of training examples.

1. **Masked language modeling.** Random corpus tokens are replaced with [MASK] and predicted with the MLM head, following distributional and cloze-style word recovery (Harris, 1954; Taylor, 1953; Devlin et al., 2019). Example: “The child ate the [MASK].” → *apple*.
2. **Replaced-token detection.** A classifier marks tokens that were replaced by plausible alternatives, following ELECTRA-style corruption discrimination (Clark et al., 2020). Example: “The cat sat on the *sky*.” marks *sky* as corrupted.

3. **Connective prediction.** A connective is replaced with [unused1] and predicted from context, reflecting discourse cohesion and explicit discourse-relation markers (Halliday and Hasan, 1976; Prasad et al., 2008). Example: “It was raining, [unused1] the family stayed inside.” → *so*.
4. **Definiteness prediction.** An article is replaced with [unused2] and classified as definite or indefinite, motivated by reference and discourse status (Gundel et al., 1993). Example: “A boy found a ball. He kicked [unused2] ball.” → *definite*.
5. **Collocation naturalness.** The model chooses the more natural phrase in a pair, motivated by habitual lexical co-occurrence (Firth, 1957). Example: *strong tea* is preferred over *powerful tea*.
6. **Function-word recovery.** Closed-class function words are masked and predicted with the MLM head, targeting syntactic structure encoded by function words (Fries, 1952). Example: “She went [MASK] school.” → *to*.
7. **Agreement prediction.** Agreement-bearing words are masked and predicted with the MLM head, targeting morphosyntactic dependencies (Corbett, 2006). Example: “The dogs [MASK] running.” → *are*.
8. **Grammar minimal-pair ranking.** A classifier chooses between generated grammatical and minimally corrupted alternatives, inspired by controlled acceptability contrasts and BLiMP-style minimal pairs (Chomsky, 1957; Warstadt et al., 2020). Example: prefer “The dogs run” over “The dogs runs.”
9. **Conceptual plausibility choice.** A classifier chooses the plausible context for a target sentence, motivated by selectional preference and EWoK-style plausibility (Resnik, 1996; Ivanova et al., 2025). Example: for “The child ate the apple,” prefer “The apple was a fruit” over “The apple was a stone.”
10. **MLM-head pair ranking.** Plausibility pairs are scored through MLM pseudo-likelihood rather than a classifier (Salazar et al., 2020). Example: “The child ate the [MASK].” ranks *apple* above *stone*.
11. **Semantic cloze ranking.** The model ranks plausible and implausible single-token completions at one masked position, combining cloze completion with plausibility contrasts (Taylor, 1953; Resnik, 1996; Ivanova et al., 2025). Example: “A bird can [MASK].” ranks *fly* above *read*.
12. **Relational semantic cloze.** The same single-mask ranking loss is applied to relation and spatial templates, motivated by spatial and relational semantics (Landau and Jackendoff, 1993; Talmy, 2000). Example: “The picture is [MASK] the wall.” ranks *on* above *under*.

B Multi-Task Fusion Diagram

Figure 3 makes explicit where the tasks remain separate and where they are combined. The model alternates task-specific batches and losses, but every selected objective updates the same encoder parameters θ . Only this shared encoder and the MLM head are retained for BabyLM evaluation.

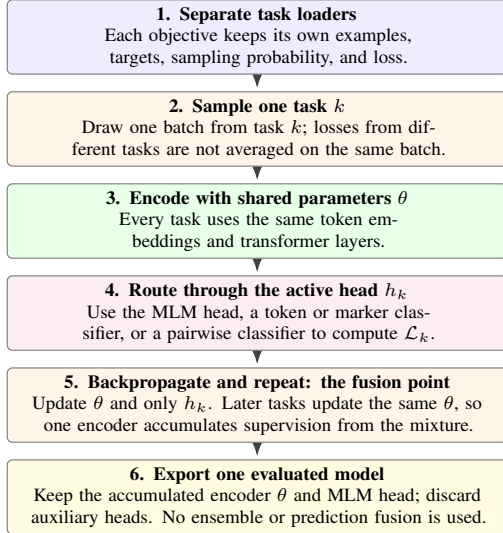


Figure 3: Multi-task fusion and export. Tasks remain separate at the data, target, head, and loss levels. They become one model because alternating task losses update the same encoder parameters.

C Result Sources

All reported scores were collected from local outputs of the official BabyLM strict fast-evaluation pipeline. For each checkpoint, the score was verified against the corresponding experiment summary artifact before being added to the paper tables. No new experiments, training, finetuning, evaluation, or checkpoint regeneration were run while preparing this draft.

D Artifact Notes

The empirical comparison treats the semantic cloze 5% final checkpoint as the EWoK-focused condition because it achieves the highest verified local official fast-evaluation EWoK score among the experimental checkpoints. All reported table values are taken from local official BabyLM fast-evaluation artifacts generated for the corresponding checkpoints. The exported artifact is a standard `AutoModelForMaskedLM` checkpoint containing the shared encoder and MLM head, without

auxiliary training heads. The model artifact is provided through the official submission system.